MWERA 2026 WORKSHOP

Expressive Python

Applying Python to Educational and Scientific Data

A hands-on, beginner-friendly workshop using Google Colab

```
>>  import pandas as pd
>   import numpy as np
    from scipy import stats
    import statsmodels
    ….. .
```

IMPORT → INSPECT → SUMMARIZE → VISUALIZE → MODEL

Educational data • Reproducible analysis • Active learning

Presenter Role

PRIMARY FACILITATOR

Graduate student in Instructional Technology pursuing an M.S.Ed specializing in Assistive Technologies, alongside a Certificate of Graduate Study in Advanced Quantitative Methodology at Northern Illinois University.

Experience includes Python for data science and hands-on design/presentation of Python-focused educational workshops for student participants and field professionals.

ACADEMIC TRAJECTORY

The facilitator plans to continue toward a Ph.D. and brings together instructional technology, quantitative methodology, data science, and teaching practice.

This workshop is structured to make technical content approachable for beginner-friendly research audiences.

Anticipated Audience & Involvement

Designed for applied researchers—students or faculty, at any level—who want to use Python to understand data in interesting ways.

BRING A LAPTOP

Participants are encouraged to run Python syntax during the workshop rather than only watch demonstrations.

FOLLOW ALONG

Handouts provide instructions for some of the basics of Python used to follow along as the workshop progresses.

ASK QUESTIONS

Dedicated time at the end supports questions about Python, analysis workflows, and next steps.

No previous knowledge of Python or Google Colab is assumed.

PURPOSE & SCOPE

Workshop Abstract

Python as an expressive research toolkit

- Python is a powerful, versatile language valued for readability, conciseness, interactivity, and statistical/data-science workflows.
- The workshop uses Google Collab to practice descriptive statistics, inferential tests, numerical operations, visualization, machine learning, and simple linear regression.
- Participants work with Pyreadstat, Pandas, NumPy, Matplotlib, SciPy, and Statsmodels, with no prior Python or Collab experience assumed.

WHY COLAB?

When you run code in a Collab notebook, execution happens on virtual machines managed by Google Cloud rather than on your local computer. This shifts intensive computation to Google-managed servers and reduces demands on the participant's machine.

BEFORE THE WORKSHOP

Python may be installed from python.org for Windows, Mac, or Linux, but Collab itself requires no local installation.

IMPORTANCE OF THE TOPIC

Why This Topic Matters

DATA-DRIVEN RESEARCH

Educational research increasingly relies on statistical analysis, visualization, reproducibility, and machine learning. Python provides one accessible, scalable language across these tasks.

LOWER THE BARRIER

Graduate students and researchers may have limited formal programming experience. A browser-based notebook reduces setup friction and supports active learning.

BUILD CAPABILITY

Python can strengthen computational thinking and problem-solving while supporting research across computer science, social science, evaluation, and education.

Workshop emphasis

- Explore student-success data.
- Connect syntax to meaningful statistical interpretation.
- Build confidence with real data, models, and visualizations.
- Encourage continued use of Python in participants' own research.

WHAT PARTICIPANTS WILL BE ABLE TO DO

Objectives, Goals & Purposes

1 IMPORTING FROM VARIOUS FILE FORMATS

Load CSV files into a Collab workflow & convert a comma-separated values (.csv) file into an SPSS (.sav) file

2 INSPECT / EXPLORE / CLEAN

Identify structure, metadata, missing values, duplicates, errors, and inconsistent formats.

3 SUMMARIZE OUR DATA

Use Pandas to condense larger datasets into interpretable statistics and grouped results.

4 VISUALIZE OUR DATA

Use NumPy, Matplotlib, and SciPy to explore and communicate patterns.

5 MACHINE LEARNING MODEL

Run statistical tests and a simple linear regression example; connect visualization to modeling.

Handouts support each activity • The session concludes with key takeaways and participant questions.

FIVE-PART FLOW

Workshop Roadmap

01

Brief introduction to Python

Google Collab + libraries

02

Inspect / Clean Data

Pandas + NumPy

03

Summarize Data

Pandas functions

04

Data Visualization

NumPy + Matplotlib

05

Simple Linear Regression

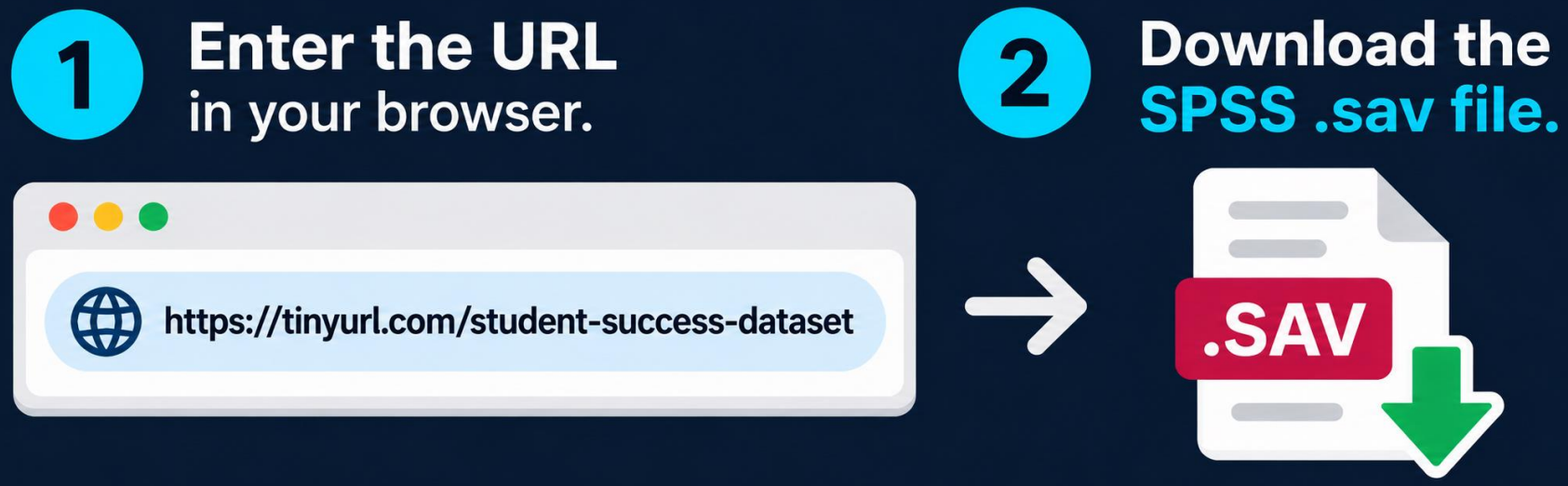
SciPy + model visualization

The workshop moves from access and file handling to interpretation, visualization, and modeling.

WRAP-UP

https://tinyurl.com/student-success-dataset

1. Enter the URL into your browser address bar.
2. Download the .SAV file.
3. Upload & Open this file in your Colab Notebook.



Google Collab: Your Cloud-Based Workspace

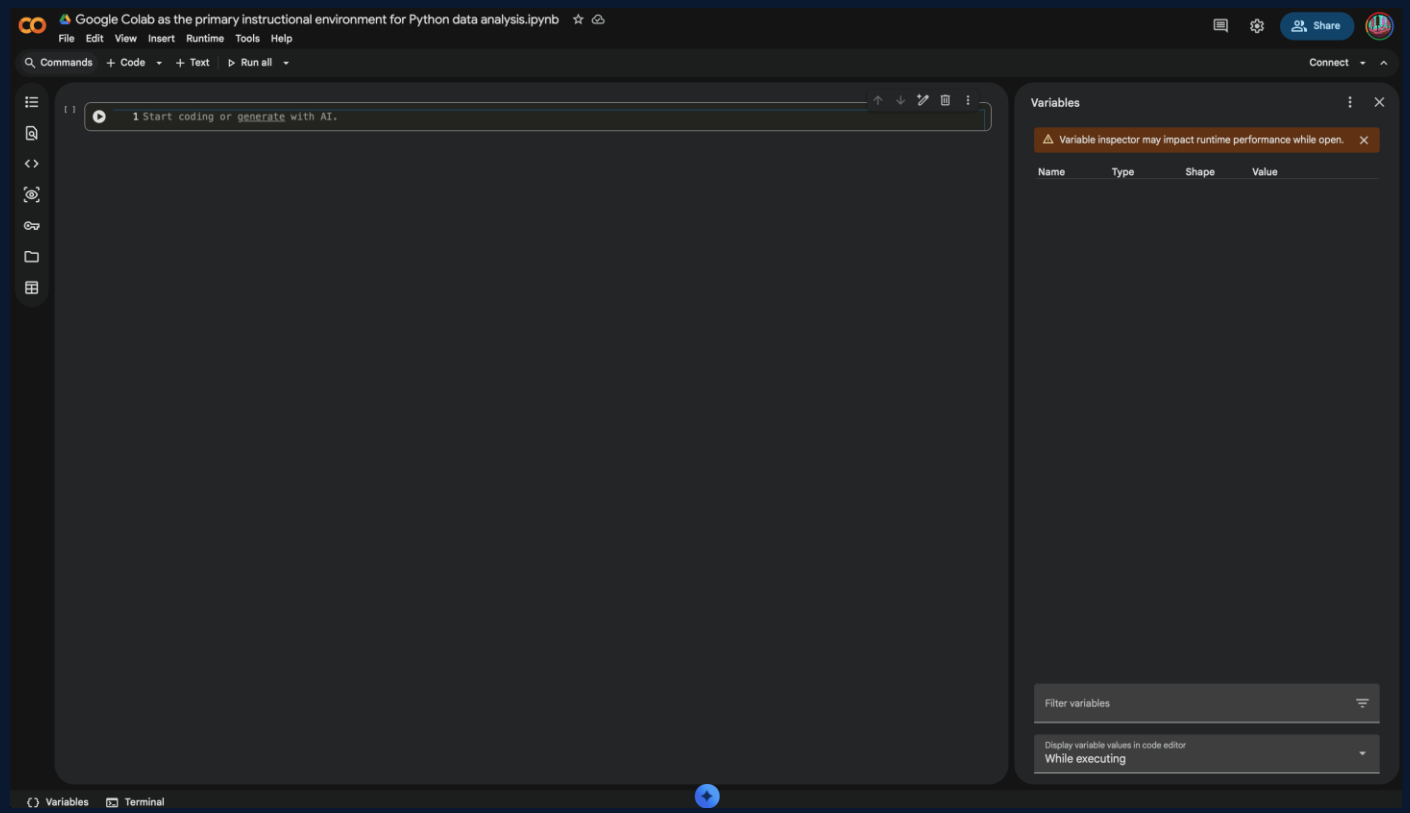


Figure 1 . Interactive Google Collab environment (colab.research.google.com)

BROWSER-BASED

Write, run, and share Python code without a time-consuming local environment setup.

CLOUD EXECUTION

Notebook code executes on Google-managed cloud virtual machines, freeing the local computer from much of the computational workload.

ACCESS

No Collab installation required.

Colab's Temporary Runtime File System

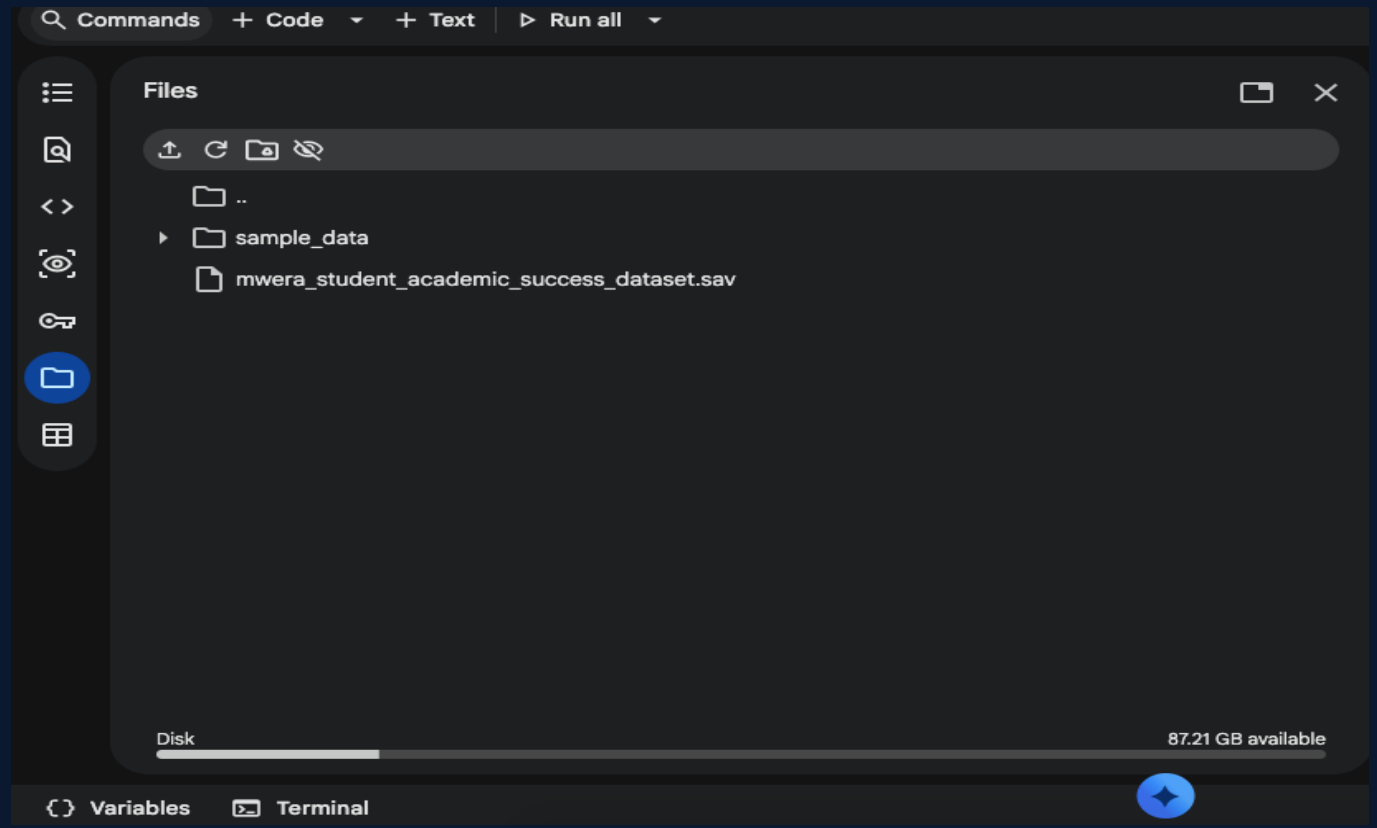


Figure 2. Files panel of Google Colab environment lets you upload files from your computer.

COLAB FILES PANEL

Temporary workspace for our notebook. When we upload the SPSS .sav dataset, it becomes available to Python within the current Colan runtime.

TEMPORARY CLOUD VM

Uploading a local file that comes right from your computer. It's being uploaded through your browser to the remote Colab virtual machine.

Collab Break-the-Ice Activity

Before analyzing data, participants will practice the core notebook actions:

01 Open the Variables menu

02 Create a new notebook

03 Rename a notebook

04 Open code cells and text cells

05 Open a terminal in a notebook

06 Restore notebooks using revision history

07 Manage sessions with Manage sessions

Install the “Must-Know” Libraries

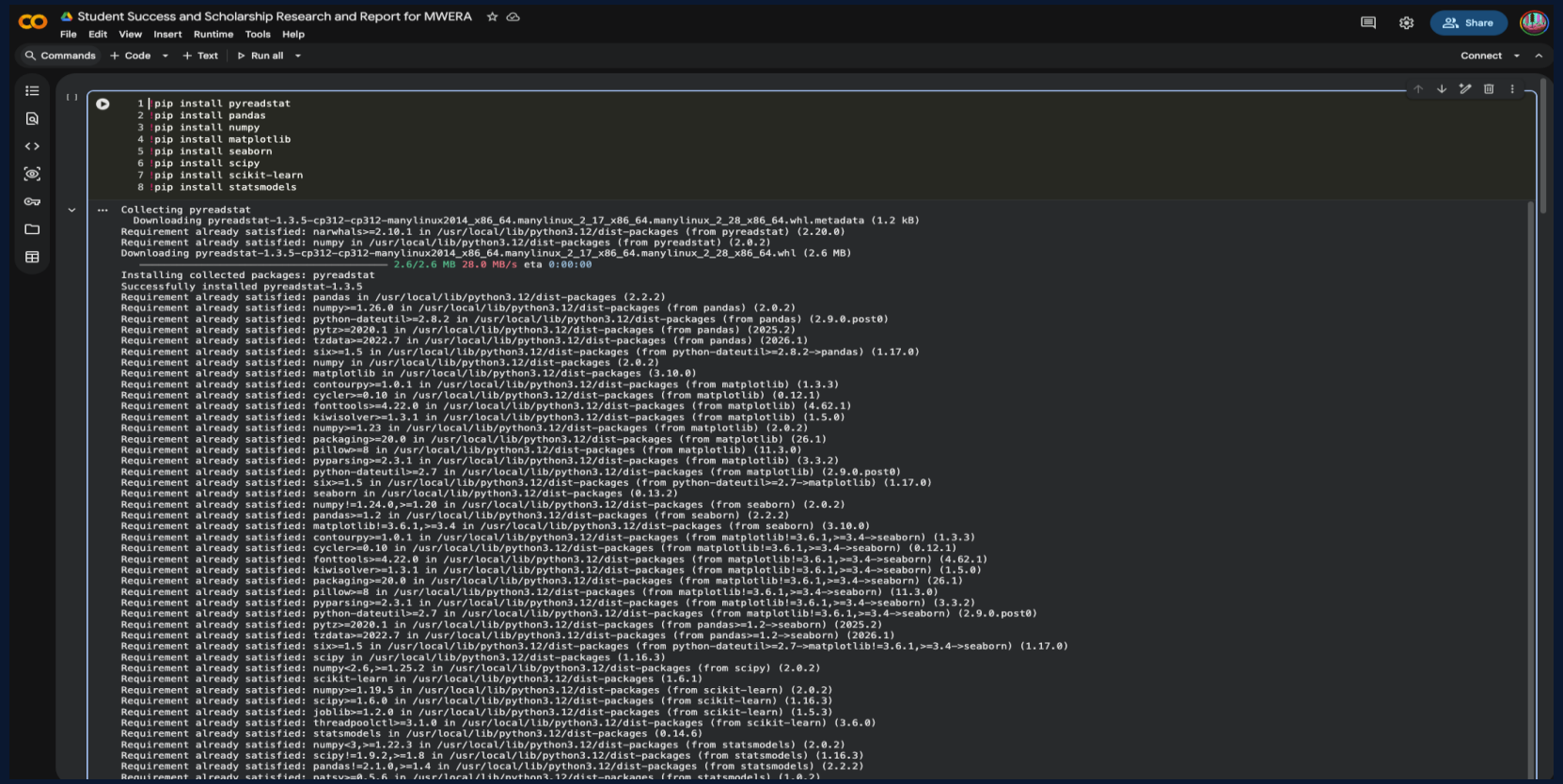


Figure 2. Installations of “must know” libraries with pip

WORKSHOP LIBRARIES

Pyreadstat — SPSS file import

Pandas — data manipulation and DataFrames

NumPy — numerical/scientific computing

Matplotlib + Seaborn — visualization

SciPy — statistical tests

Statsmodels — statistical modeling

Importing Research Data

```
[7]
1 import pandas as pd
2
3 # Essential library import (works very well for SPSS/Strata/SAS files)
4 import pyreadstat
5
6 # Key function example
7 df, meta = pyreadstat.read_sav("student_academic_success_dataset.sav")
8 df.head()
```

	student_id	age	gener	race_ethnicity	major	gpa	study_hours	attendance_rate	num_courses	first_gen	scholarship	graduation_intent
0	100001.0	21.0	Male	Asian	Undeclared	2.89	17.0	94.9	4.0	1.0	1.0	4.0
1	100002.0	21.0	Male	Native American or Alaska Native	Engineering	3.72	22.2	100.0	2.0	0.0	0.0	5.0
2	100003.0	18.0	Female	Asian	Sociology	3.02	18.2	88.8	5.0	0.0	0.0	4.0
3	100004.0	25.0	Male	Hispanic or Latino	Education	2.38	22.3	87.6	3.0	0.0	1.0	3.0
4	100005.0	23.0	Male	Black or African American	Education	2.69	23.0	87.1	4.0	0.0	0.0	5.0

Figure 3. Importing SPSS '.sav' files with pyreadstat

SPSS WORKFLOW

pyreadstat.read_sav(...) returns the data plus metadata, supporting variables and value labels common in educational datasets.

Inspect & Clean Data

Pandas and NumPy help detect and correct inaccurate records so the dataset is reliable before analysis.

COMMON TASKS

Handle missing values
Remove duplicate records
Correct errors
Standardize formats

DESCRIPTIVE CHECKS

Mean
Median
Mode
Distribution checks

WHY IT MATTERS

Cleaning connects directly to later analysis. Imputation—such as mean, median, or mode replacement—can preserve records when appropriate.

Pandas DataFrame

A two-dimensional labeled data structure whose columns can contain different data types—the central structure used throughout the workshop.

Inspect the DataFrame

```
1 # Data cleaning/inspecting tasks using Pandas library:
2
3 df.head()
```

	student_id	age	gener	race_ethnicity	major	gpa	study_hours	attendance_rate	num_courses	first_gen	scholarship	graduation_intent
0	100001	21	Male	Asian	Undeclared	2.89	17.0	94.9	4	1	1	Likely
1	100002	21	Male	Native American or Alaska Native	Engineering	3.72	22.2	100.0	2	0	0	Very likely
2	100003	18	Female	Asian	Sociology	3.02	18.2	88.8	5	0	0	Likely
3	100004	25	Male	Hispanic or Latino	Education	2.38	22.3	87.6	3	0	1	Unsure
4	100005	23	Male	Black or African American	Education	2.69	23.0	87.1	4	0	0	Very likely

Figure 5. Data manipulation using Pandas library (df.head())

THREE FAST QUESTIONS

What does the data look like?
How large is it?
What variables are available?

```
[11]
0s
1 df.shape
(250, 14)
```

Figure 6. Determine rows and columns (df.shape)

```
[15]
0s
1 df.columns
Index(['student_id', 'age', 'gener', 'race_ethnicity', 'major', 'gpa',
       'study_hours', 'attendance_rate', 'num_courses', 'first_gen',
       'scholarship', 'graduation_intent', 'sat_score', 'employed'],
      dtype='object')
```

Figure 7. Determine column names (df.columns)

Read the Metadata

```
[17]
0s
1 meta.column_names
['student_id',
 'age',
 'gener',
 'race_ethnicity',
 'major',
 'gpa',
 'study_hours',
 'attendance_rate',
 'num_courses',
 'first_gen',
 'scholarship',
 'graduation_intent',
 'sat_score',
 'employed']
```

Figure 8. Dataset metadata: meta.column_names

```
[18]
0s
1 meta.variable_value_labels
{'first_gen': {0.0: 'No', 1.0: 'Yes'},
 'scholarship': {0.0: 'No', 1.0: 'Yes'},
 'graduation_intent': {1.0: 'Very unlikely',
  2.0: 'Unlikely',
  3.0: 'Unsure',
  4.0: 'Likely',
  5.0: 'Very likely'},
 'employed': {0.0: 'No', 1.0: 'Yes'}}
```

Figure 9. meta.variable_value_labels maps coded values to human-readable labels

RESEARCH VALUE

Metadata preserves the meaning behind coded variables. For SPSS files, value-label dictionaries help translate numeric categories into interpretable labels before summarizing or modeling.

Summarize Data with Pandas

The goal is to turn a larger dataset into digestible representations that reveal averages, totals, counts, and trends.

df.head() preview rows	**df.info()** structure + types	**df.describe()** descriptive statistics	**df.value_counts()** frequency counts
df.groupby() grouped summaries	**df.sort_values()** ordered records	**df.rename()** clearer variable names	**df['new_var']** create/access a variable

Descriptive Statistics & Frequencies

```
[ ]  1 df.describe()
```

	student_id	age	gpa	study_hours	attendance_rate	num_courses	first_gen	scholarship	sat_score	employed
count	250.000000	250.00000	250.000000	250.000000	250.00000	250.00000	250.000000	250.0000	250.000000	250.000000
mean	100125.500000	21.54400	3.162960	21.305600	88.07880	4.38800	0.364000	0.2800	1147.720000	0.456000
std	72.312977	2.92477	0.441305	5.868065	7.84018	0.90816	0.482114	0.4499	146.380705	0.499059
min	100001.000000	18.00000	1.750000	6.100000	67.80000	2.00000	0.000000	0.0000	820.000000	0.000000
25%	100063.250000	19.00000	2.852500	17.400000	82.90000	4.00000	0.000000	0.0000	1060.000000	0.000000
50%	100125.500000	21.00000	3.165000	21.600000	87.95000	4.00000	0.000000	0.0000	1145.000000	0.000000
75%	100187.750000	24.00000	3.480000	25.300000	94.40000	5.00000	1.000000	1.0000	1240.000000	1.000000
max	100250.000000	32.00000	4.000000	38.000000	100.00000	6.00000	1.000000	1.0000	1520.000000	1.000000

Figure 10. df.describe() displays descriptive statistics for numerical data types

```
[21]  1 df.value_counts()
```

student_id	age	gender	race_ethnicity	major	gpa	study_hours	attendance_rate	num_courses	first_gen	scholarship	graduation_intent	sat_score	employed	count
100001	21	Male	Asian	Undeclared	2.89	17.0	94.9	4	1	1	Likely	1320	0	1
100002	21	Male	Native American or Alaska Native	Engineering	3.72	22.2	100.0	2	0	0	Very likely	1250	1	1
100003	18	Female	Asian	Sociology	3.02	18.2	88.8	5	0	0	Likely	1100	0	1
100004	25	Male	Hispanic or Latino	Education	2.38	22.3	87.6	3	0	1	Unsure	830	1	1
100005	23	Male	Black or African American	Education	2.69	23.0	87.1	4	0	0	Very likely	1030	1	1
...	...	...	...	...	...	...	...	...	...	...	...	...	...	...
100246	18	Male	White	Computer Science	3.49	34.4	87.6	4	0	1	Very likely	1260	0	1
100247	19	Male	White	Education	3.70	29.5	89.7	5	1	0	Very likely	1160	0	1
100248	23	Female	Black or African American	Computer Science	3.64	33.8	100.0	5	0	0	Very likely	1210	0	1
100249	19	Female	White	Engineering	3.26	22.6	100.0	5	0	0	Unsure	980	0	1
100250	24	Nonbinary	White	Computer Science	3.33	25.8	100.0	4	0	1	Very likely	1280	1	1

250 rows × 1 columns

dtype: int64

Figure 11. df.value_counts() displays the frequency of unique rows

DESCRIBE

Quickly inspect count, mean, standard deviation, quartiles, minimum, and maximum for numerical variables.

VALUE COUNTS

Count repeated values or combinations to identify common categories and patterns in the DataFrame.

Group, Compare, Interpret

```
[30]
✓ 0s
    1 mean_gpa = df.groupby('study_hours')['gpa'].mean()
    2 print(mean_gpa)

... study_hours
    6.1     3.08
    7.8     2.86
    8.6     1.75
    9.1     2.98
    9.4     2.75
            ...
    33.8    3.64
    34.4    3.49
    35.1    3.94
    35.9    2.93
    38.0    3.35
    Name: gpa, Length: 160, dtype: float64
```

Figure 12. Displaying mean_gpa using the df.groupby() function

THE QUESTION

How does mean GPA vary across different values of study_hours?

THE PATTERN

Grouping lets researchers summarize an outcome within meaningful categories or values before moving to visualization and modeling.

Summaries are not the endpoint—they are the bridge from raw records to research questions.

NumPy + Data Visualization

NUMPY

Core scientific-computing package for large multidimensional arrays and matrices. Workshop functions include np.where(), np.log(), np.mean(), and np.std().

MATPLOTLIB

Matplotlib: plt.scatter(), plt.hist(), plt.boxplot(), plt.bar()

Visualization supports interpretation, exploration, and communication of patterns in educational and scientific datasets.

NUMERICAL OPERATIONS → SUMMARY → VISUAL PATTERN → STATISTICAL MODEL

Standard Deviation in Context

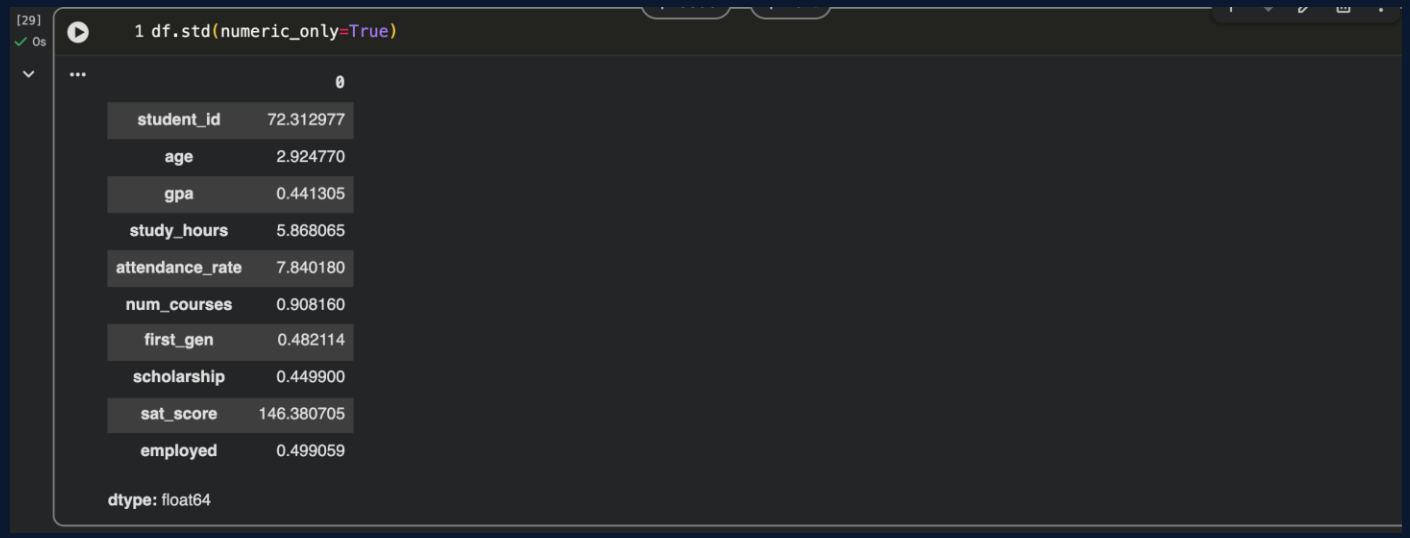


Figure 13. np.std() demonstrating standard deviation across numeric variables

WHAT IT DOES

np.std() measures dispersion: how far values tend to spread around their mean.

WORKSHOP CONNECTION

Use numerical summaries to identify scale, variability, and possible issues before visualizing or modeling.

See the Relationship

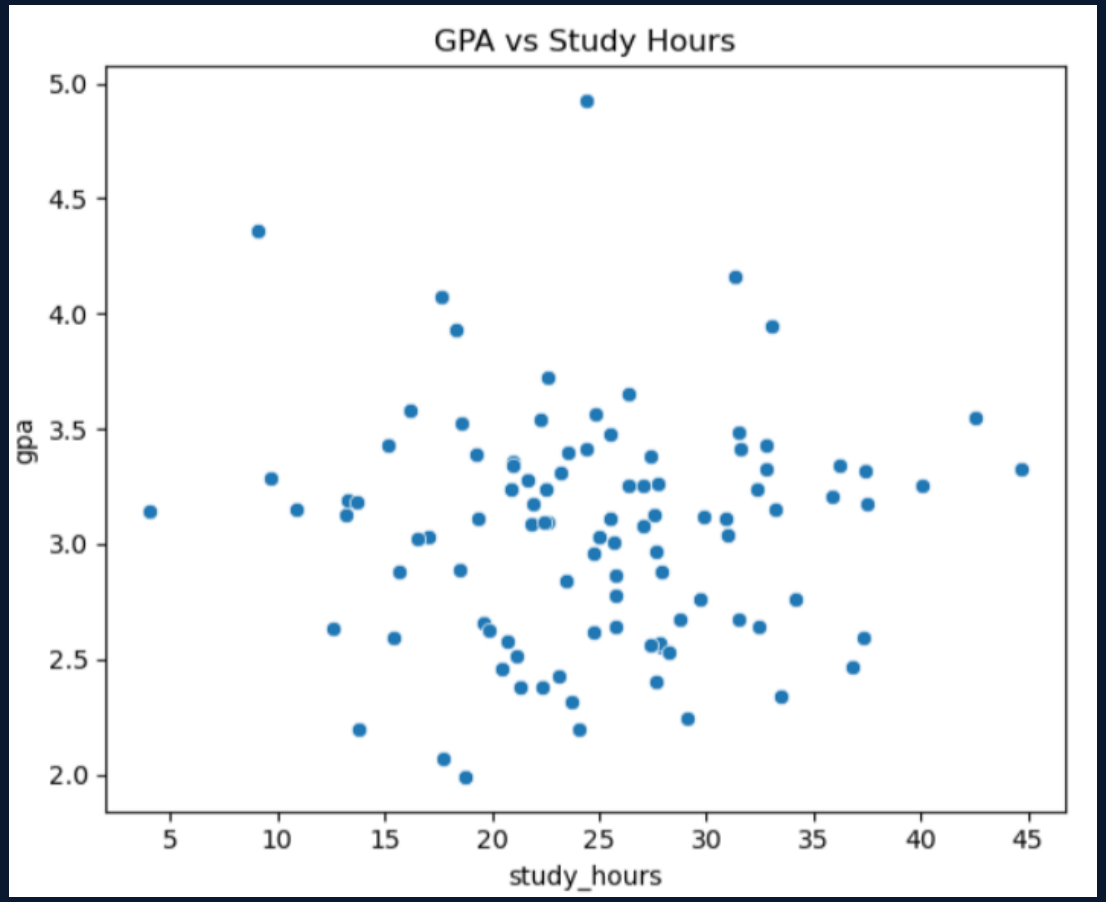


Figure 14. sns.scatterplot() — GPA vs. Study Hours

X

Study hours

Y

GPA

WHY PLOT FIRST?

Visualization reveals shape, spread, outliers, and the plausibility of a relationship before formal modeling.

From Visualization to Statistical Testing

Linear regression is presented as the bridge where data visualization meets statistical modeling.

SIMPLE LINEAR REGRESSION

Participants see how Python syntax can connect a predictor and outcome, generate predictions, and visualize model performance.

`Question → test/model → output → interpretation → visualization`

Visualize Model Performance

```
[45]
# Below is a vizualization of the performance of the linear regression model.

import matplotlib.pyplot as plt
import seaborn as sns

# We accomplish this by creating scatter plot of actual vs. predicted values
plt.figure(figsize=(8, 6))
sns.scatterplot(x=y_test, y=y_pred, alpha=0.6)
plt.plot([y_test.min(), y_test.max()], [y_test.min(), y_test.max()], 'r--', lw=2) # Add a diagonal line for reference
plt.title('Actual vs. Predicted Study Hours (Linear Regression)')
plt.xlabel('Actual Study Hours')
plt.ylabel('Predicted Study Hours')
plt.grid(True, linestyle='--', alpha=0.7)
plt.tight_layout()
plt.show()
```

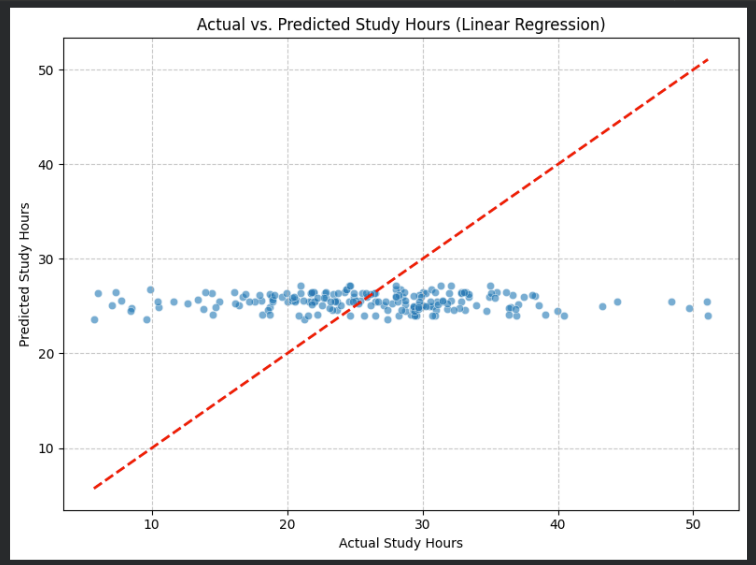


Figure 15. Scatter plot visualizing the performance of a linear regression model

ACTUAL

Observed study-hours values in the dataset.

PREDICTED

Values generated by the fitted linear regression model.

INTERPRET

Compare observed and predicted values to discuss fit, residuals, and model usefulness.

WHAT TO LEAVE WITH

Key Takeaways

ACCESS

Colab lowers setup barriers and moves execution to cloud-based compute.

STRUCTURE

Pandas and Pyreadstat make educational data and SPSS metadata easier to inspect.

NUMERACY

NumPy supports efficient numerical operations and foundational scientific computing.

COMMUNICATION

Matplotlib and Seaborn turn numerical patterns into interpretable visual evidence.

INFERENCE

SciPy and Statsmodels extend the workflow into testing and statistical modeling.

TRANSFER

The same workflow can be adapted to participants' own educational and scientific research questions.

Questions + Next Steps
Scan the QR Code & Take a Survey



What will you analyze with Python next?

Suggested next step: open a Colab notebook, import a small dataset, inspect it, summarize one variable, visualize one relationship, and run one statistical test or model.

REFERENCE

Rathod, D. P. R. (2025). Python's Potential to Enhance Learning in Higher and Distance Education. 12(3).

Thank you